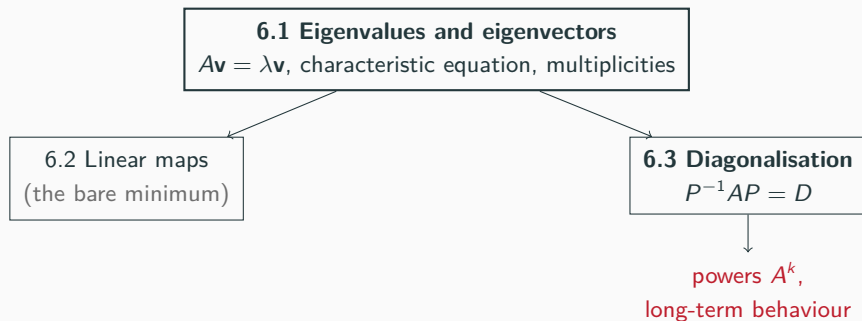


Linear Algebra

6 Eigenvalues and eigenvectors

José Firmino Aguilar Madeira



Uses: complex numbers (Ch. 1), subspaces and dimension (Ch. 2), systems and rank (Ch. 3), determinants (Ch. 4), orthogonality (Ch. 5).

Exercises: Sheet 6, Ex. 99–102 (numbering continues from Sheet 5).

6.1 Eigenvalues and eigenvectors

A matrix usually changes the direction of vectors

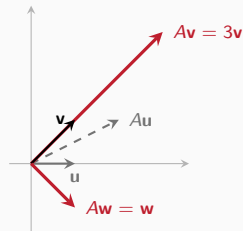
With $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$:

$$A \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad A \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad A \begin{bmatrix} 1 \\ -1 \end{bmatrix} = 1 \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

The first vector changes direction.

The other two **keep their direction**: they are just multiplied by a number.

Question for this chapter: which vectors keep their direction, and by what number are they multiplied?



Definition

Let $\mathbb{K} = \mathbb{R}$ or $\mathbb{K} = \mathbb{C}$ and let A be a square matrix of order n with entries in $\mathbb{K}$.

Definition (eigenvalue, eigenvector)

$\lambda \in \mathbb{K}$ is an **eigenvalue** of A if there exists $\mathbf{v} \in \mathbb{K}^n$ such that

$$\mathbf{v} \neq \mathbf{0} \quad \text{and} \quad A\mathbf{v} = \lambda\mathbf{v}.$$

Each such $\mathbf{v}$ is an **eigenvector** of A associated with λ .

- The condition $\mathbf{v} \neq \mathbf{0}$ is essential: $A\mathbf{0} = \lambda\mathbf{0}$ for *any* λ .
- An eigenvalue may be 0; it is the eigenvector that cannot be zero.
- Vectors in $\mathbb{K}^n$ are written as columns; in the text, $(x_1, \dots, x_n)$ denotes that column.

Also called characteristic value/vector.

How to find the eigenvalues

$$A\mathbf{v} = \lambda\mathbf{v} \iff (A - \lambda I_n)\mathbf{v} = \mathbf{0}.$$

This is a **homogeneous system**. There are eigenvectors associated with λ iff it has a non-zero solution:

$$\text{infinitely many solutions} \iff \text{car}(A - \lambda I_n) < n \iff \det(A - \lambda I_n) = 0.$$

(Chs. 3 and 4: for a square matrix, rank $n \iff$ determinant $\neq 0$.)

Theorem

Let A be square of order n over $\mathbb{K}$. Then $\lambda \in \mathbb{K}$ is an eigenvalue of A iff

$$\det(A - \lambda I_n) = 0.$$

$p(\lambda) = \det(A - \lambda I_n)$: **characteristic polynomial** (degree n); $p(\lambda) = 0$: **characteristic equation**.

Method

1. Compute $p(\lambda) = \det(A - \lambda I_n)$.
2. Solve $p(\lambda) = 0$ in $\mathbb{K}$: the roots are the eigenvalues.
3. For each eigenvalue λ , solve the homogeneous system

$$(A - \lambda I_n) \mathbf{v} = \mathbf{0}$$

by Gaussian elimination. The **non-zero** solutions are the eigenvectors associated with λ .

A check that costs nothing

At the end, confirm $A\mathbf{v} = \lambda\mathbf{v}$ for each eigenvector found.

Example 1: eigenvalues

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \quad (\text{the matrix from the opening figure}).$$

$$p(\lambda) = \det \begin{bmatrix} 2 - \lambda & 1 \\ 1 & 2 - \lambda \end{bmatrix} = (2 - \lambda)^2 - 1 = \lambda^2 - 4\lambda + 3 = (\lambda - 1)(\lambda - 3).$$

Example 1: eigenvalues

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \quad (\text{the matrix from the opening figure}).$$

$$p(\lambda) = \det \begin{bmatrix} 2 - \lambda & 1 \\ 1 & 2 - \lambda \end{bmatrix} = (2 - \lambda)^2 - 1 = \lambda^2 - 4\lambda + 3 = (\lambda - 1)(\lambda - 3).$$

Eigenvalues: $\lambda_1 = 1$ and $\lambda_2 = 3$.

Example 1: eigenvectors

$$\lambda = 3: \quad A - 3I_2 = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \xrightarrow{L_2 \rightarrow L_2 + L_1} \begin{bmatrix} -1 & 1 \\ 0 & 0 \end{bmatrix} \Rightarrow -x_1 + x_2 = 0 \iff x_2 = x_1.$$

$$\mathbf{v} = x_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad x_1 \neq 0.$$

Example 1: eigenvectors

$$\lambda = 3: \quad A - 3I_2 = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \xrightarrow{L_2 \rightarrow L_2 + L_1} \begin{bmatrix} -1 & 1 \\ 0 & 0 \end{bmatrix} \Rightarrow -x_1 + x_2 = 0 \iff x_2 = x_1.$$

$$\mathbf{v} = x_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad x_1 \neq 0.$$

$$\lambda = 1: \quad A - I_2 = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \xrightarrow{L_2 \rightarrow L_2 - L_1} \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \Rightarrow x_2 = -x_1.$$

$$\mathbf{v} = x_1 \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \quad x_1 \neq 0.$$

Example 1: eigenvectors

$$\lambda = 3: \quad A - 3I_2 = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \xrightarrow{L_2 \rightarrow L_2 + L_1} \begin{bmatrix} -1 & 1 \\ 0 & 0 \end{bmatrix} \Rightarrow -x_1 + x_2 = 0 \iff x_2 = x_1.$$

$$\mathbf{v} = x_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad x_1 \neq 0.$$

$$\lambda = 1: \quad A - I_2 = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \xrightarrow{L_2 \rightarrow L_2 - L_1} \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \Rightarrow x_2 = -x_1.$$

$$\mathbf{v} = x_1 \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \quad x_1 \neq 0.$$

Check: $A(1, 1) = (3, 3) = 3(1, 1)$ and $A(1, -1) = (1, -1)$.

Eigenspace and multiplicities

Definitions

Let λ be an eigenvalue of A (order n).

- **Eigenspace:** $E(\lambda) = \{\mathbf{x} \in \mathbb{K}^n : (A - \lambda I_n)\mathbf{x} = \mathbf{0}\}$.
It is a subspace of $\mathbb{K}^n$ (Ch. 2); the eigenvectors are its *non-zero* elements.
- **Algebraic multiplicity** $m_a(\lambda)$: multiplicity of λ as a root of p .
- **Geometric multiplicity** $m_g(\lambda) = \dim E(\lambda) = n - \text{car}(A - \lambda I_n)$:
maximum number of lin. indep. eigenvectors associated with λ (the number of free variables g of the system, Ch. 3).

Theorem

For each eigenvalue λ of A : $1 \leq m_g(\lambda) \leq m_a(\lambda)$.

In Example 1: $m_a = m_g = 1$ for $\lambda = 1$ and for $\lambda = 3$.

Example 2: the two multiplicities can differ

$$B = \begin{bmatrix} 5 & 3 \\ 0 & 5 \end{bmatrix}. \quad p(\lambda) = \det \begin{bmatrix} 5 - \lambda & 3 \\ 0 & 5 - \lambda \end{bmatrix} = (5 - \lambda)^2.$$

Example 2: the two multiplicities can differ

$$B = \begin{bmatrix} 5 & 3 \\ 0 & 5 \end{bmatrix}. \quad p(\lambda) = \det \begin{bmatrix} 5 - \lambda & 3 \\ 0 & 5 - \lambda \end{bmatrix} = (5 - \lambda)^2.$$

Only eigenvalue: $\lambda = 5$, with $m_a(5) = 2$.

Example 2: the two multiplicities can differ

$$B = \begin{bmatrix} 5 & 3 \\ 0 & 5 \end{bmatrix}. \quad p(\lambda) = \det \begin{bmatrix} 5 - \lambda & 3 \\ 0 & 5 - \lambda \end{bmatrix} = (5 - \lambda)^2.$$

Only eigenvalue: $\lambda = 5$, with $m_a(5) = 2$.

$$B - 5I_2 = \begin{bmatrix} 0 & 3 \\ 0 & 0 \end{bmatrix} \Rightarrow 3x_2 = 0 \iff x_2 = 0, \quad \mathbf{v} = x_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad x_1 \neq 0.$$

$\text{car}(B - 5I_2) = 1$, so $m_g(5) = 2 - 1 = 1 < m_a(5)$.

We shall see in 6.3 that this is what prevents B from being diagonalisable.

Properties

Let A be square of order n and let $\lambda_1, \dots, \lambda_n$ be its eigenvalues in $\mathbb{C}$, each repeated m_a times (there are n of them, by the Fundamental Theorem of Algebra, Ch. 1). Then

$$\lambda_1 + \dots + \lambda_n = \operatorname{tr} A, \quad \lambda_1 \cdots \lambda_n = \det A.$$

- Example 1: $1 + 3 = 4 = \operatorname{tr} A$ and $1 \cdot 3 = 3 = \det A$.
- Consequence: A is invertible $\iff 0$ is not an eigenvalue of A .
- If A is **triangular**, the eigenvalues are the entries on the main diagonal:
 $A - \lambda I_n$ is triangular, so $p(\lambda) = (a_{11} - \lambda) \cdots (a_{nn} - \lambda)$ (Ch. 4, VIII). (Example 2: $\lambda = 5$.)

They help catch arithmetic errors; they do not replace the calculation.

Ex. 99 Eigenvalues and eigenvectors, with their multiplicities. In class, five items covering the different cases:

- a) two distinct real eigenvalues;
- b) no real eigenvalues;
- i) repeated eigenvalue with $m_g = m_a$;
- f) repeated eigenvalue with $m_g < m_a$;
- o) triangular matrix (diagonal shortcut).

The remaining items are for independent study.

Solutions on the solutions sheet (Moodle), made available after the class.

6.2 Linear maps

A linear map and its matrix

Definition

$T : \mathbb{K}^n \rightarrow \mathbb{K}^m$ is a **linear map** if, for all $\mathbf{u}, \mathbf{v} \in \mathbb{K}^n$ and $\alpha \in \mathbb{K}$,

$$T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v}), \quad T(\alpha \mathbf{u}) = \alpha T(\mathbf{u}).$$

If $m = n$, it is called a **linear operator**.

Matrix in the standard basis: column j is $T(\mathbf{e}_j)$. Then $T(\mathbf{x}) = A\mathbf{x}$ for every $\mathbf{x}$.

Example: $T(x, y) = (5x + 3y, 5y)$. $T(1, 0) = (5, 0)$, $T(0, 1) = (3, 5) \Rightarrow A = \begin{bmatrix} 5 & 3 \\ 0 & 5 \end{bmatrix}$.

This is the matrix B from Example 2.

Eigenvalues of a linear operator

λ is an eigenvalue of $T : \mathbb{K}^n \rightarrow \mathbb{K}^n$ if $T(\mathbf{v}) = \lambda\mathbf{v}$ for some $\mathbf{v} \neq \mathbf{0}$. The set of eigenvalues is the **spectrum** of T .

- They are computed from the matrix of T : they are the eigenvalues of A .
- In another basis, the matrix of T becomes $P^{-1}AP$ (P invertible). The eigenvalues do not change:

$$\det(P^{-1}AP - \lambda I) = \det(P^{-1}(A - \lambda I)P) = \det(P^{-1}) \det(A - \lambda I) \det P = \det(A - \lambda I).$$

(Determinant of a product, Ch. 4.)

A and $P^{-1}AP$ are called *similar*: they have the same characteristic polynomial.

Example: $T(x, y) = (5x + 3y, 5y)$ has spectrum $\{5\}$; the eigenvectors are $(x, 0)$, $x \neq 0$.

On Sheet 6: Ex. 100 (operator F) and Ex. 101 (operator G , which is Example 3 below).

Example 3: in $\mathbb{R}$ or in $\mathbb{C}$?

$G : \mathbb{R}^3 \rightarrow \mathbb{R}^3$, $G(x, y, z) = (y, -x, z)$ (rotation by -90° about the Oz axis).

$$A_G = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad p(\lambda) = (1 - \lambda)(\lambda^2 + 1).$$

In $\mathbb{R}$: $\lambda^2 + 1$ has no real roots.

Spectrum $\{1\}$; eigenvectors $(0, 0, z)$, $z \neq 0$.

The axis of rotation is the only direction that is kept.

In $\mathbb{C}$: spectrum $\{1, i, -i\}$.

$\lambda = i$: $(1, i, 0)$;

$\lambda = -i$: $(1, -i, 0)$ (and non-zero multiples).

The 2×2 block is the rotation R_{-90° (Ch. 3); R_θ has eigenvalues $\text{cis}(\pm\theta)$ (Ch. 1).

Course convention

Unless stated otherwise, we work in $\mathbb{K} = \mathbb{R}$. When $p(\lambda)$ has complex roots, this must be stated in the answer.

6.3 Diagonalisation

Definition

The matrix A , square of order n over $\mathbb{K}$, is **diagonalisable** (in $\mathbb{K}$) if there exists an invertible P , of order n over $\mathbb{K}$, such that

$$P^{-1}AP = D \quad \text{is a diagonal matrix.}$$

What is it for? $A = PDP^{-1}$, so

$$A^k = (PDP^{-1})(PDP^{-1}) \cdots (PDP^{-1}) = P D^k P^{-1},$$

and D^k is computed entry by entry. (See the powers example later on.)

When is A diagonalisable?

Theorem 1

Eigenvectors associated with pairwise *distinct* eigenvalues are linearly independent.

Theorem 2 (criterion)

Let A be square of order n over $\mathbb{K}$, with distinct eigenvalues $\lambda_1, \dots, \lambda_r$ in $\mathbb{K}$. The following are equivalent:

1. A is diagonalisable in $\mathbb{K}$;
2. A has n linearly independent eigenvectors;
3. $m_g(\lambda_1) + \dots + m_g(\lambda_r) = n$.

In that case, $P = [\mathbf{v}_1 \ \dots \ \mathbf{v}_n]$ has as columns n lin. indep. eigenvectors and $D = \text{diag}(\lambda_{(1)}, \dots, \lambda_{(n)})$ has the corresponding eigenvalues, **in the same order**.

A sufficient (not necessary) condition

Corollary

If A , of order n , has n *distinct* eigenvalues in $\mathbb{K}$, then A is diagonalisable in $\mathbb{K}$.

Each $m_g(\lambda_i) \geq 1$ and there are n eigenvalues, so the sum is n .

Note

- The condition is **sufficient, not necessary**: Example 4 is diagonalisable with a repeated eigenvalue.
- A repeated eigenvalue does *not* prevent diagonalisation. What matters is whether $m_g(\lambda) = m_a(\lambda)$, i.e. whether the sum of the **geometric** multiplicities reaches n .

Example 4: eigenvalues

$$C = \begin{bmatrix} 2 & 0 & 0 \\ -6 & 2 & -3 \\ -6 & 0 & -1 \end{bmatrix}. \quad \text{Expanding along the 1st row (Laplace):}$$

$$p(\lambda) = \det \begin{bmatrix} 2 - \lambda & 0 & 0 \\ -6 & 2 - \lambda & -3 \\ -6 & 0 & -1 - \lambda \end{bmatrix} = (2 - \lambda) \det \begin{bmatrix} 2 - \lambda & -3 \\ 0 & -1 - \lambda \end{bmatrix} = (2 - \lambda)^2(-1 - \lambda).$$

Example 4: eigenvalues

$$C = \begin{bmatrix} 2 & 0 & 0 \\ -6 & 2 & -3 \\ -6 & 0 & -1 \end{bmatrix}. \quad \text{Expanding along the 1st row (Laplace):}$$

$$p(\lambda) = \det \begin{bmatrix} 2 - \lambda & 0 & 0 \\ -6 & 2 - \lambda & -3 \\ -6 & 0 & -1 - \lambda \end{bmatrix} = (2 - \lambda) \det \begin{bmatrix} 2 - \lambda & -3 \\ 0 & -1 - \lambda \end{bmatrix} = (2 - \lambda)^2(-1 - \lambda).$$

Eigenvalues: $\lambda = 2$ with $m_a(2) = 2$ and $\lambda = -1$ with $m_a(-1) = 1$.

Check: $2 + 2 + (-1) = 3 = \text{tr } C$ and $2 \cdot 2 \cdot (-1) = -4 = \det C$.

C is this course's **running matrix**: its columns $\mathbf{c}_1, \mathbf{c}_2, \mathbf{c}_3$ (Ch. 2), $\text{car } C = 3$ and C^{-1} (Ch. 3), $\det C = -4$ (Ch. 4). Now: -4 is the product of the eigenvalues.

Example 4: eigenvectors

$$\lambda = 2: \quad C - 2I_3 = \begin{bmatrix} 0 & 0 & 0 \\ -6 & 0 & -3 \\ -6 & 0 & -3 \end{bmatrix} \longrightarrow \begin{bmatrix} 2 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \Rightarrow \quad x_3 = -2x_1 \quad (x_1, x_2 \text{ free}).$$

$$\mathbf{v} = x_1 \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad (x_1, x_2) \neq (0, 0). \quad \text{car}(C - 2I_3) = 1 \Rightarrow m_g(2) = 2.$$

Example 4: eigenvectors

$$\lambda = 2: \quad C - 2I_3 = \begin{bmatrix} 0 & 0 & 0 \\ -6 & 0 & -3 \\ -6 & 0 & -3 \end{bmatrix} \longrightarrow \begin{bmatrix} 2 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow x_3 = -2x_1 \quad (x_1, x_2 \text{ free}).$$

$$\mathbf{v} = x_1 \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad (x_1, x_2) \neq (0, 0). \quad \text{car}(C - 2I_3) = 1 \Rightarrow m_g(2) = 2.$$

$$\lambda = -1: \quad C + I_3 = \begin{bmatrix} 3 & 0 & 0 \\ -6 & 3 & -3 \\ -6 & 0 & 0 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow x_1 = 0, \quad x_2 = x_3.$$

$$\mathbf{v} = x_3 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \quad x_3 \neq 0. \quad m_g(-1) = 1.$$

Example 4: diagonalisation

$m_g(2) + m_g(-1) = 2 + 1 = 3 = n$, so C is diagonalisable (Theorem 2).

Example 4: diagonalisation

$m_g(2) + m_g(-1) = 2 + 1 = 3 = n$, so C is diagonalisable (Theorem 2).

$$P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ -2 & 0 & 1 \end{bmatrix}, \quad D = P^{-1}CP = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -1 \end{bmatrix}.$$

Example 4: diagonalisation

$m_g(2) + m_g(-1) = 2 + 1 = 3 = n$, so C is diagonalisable (Theorem 2).

$$P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ -2 & 0 & 1 \end{bmatrix}, \quad D = P^{-1}CP = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -1 \end{bmatrix}.$$

- The columns of P are the eigenvectors $(1, 0, -2)$, $(0, 1, 0)$, $(0, 1, 1)$; the diagonal of D has $2, 2, -1$, **in the same order**.
- Check without inverting P : $CP = PD$. ($\det P = 1$, so P is invertible.)
- Swapping the order of the columns of P swaps the order of the diagonal of D .

When it is not diagonalisable

Example 2: $B = \begin{bmatrix} 5 & 3 \\ 0 & 5 \end{bmatrix}$. $m_g(5) = 1 < 2 = n \Rightarrow$ not diagonalisable.

Example 3: A_G (rotation by -90° about Oz).

- In $\mathbb{R}$: there is only the eigenvalue 1, with $m_g(1) = 1 < 3 \Rightarrow$ not diagonalisable in $\mathbb{R}$.
- In $\mathbb{C}$: $1, i, -i$ are 3 distinct eigenvalues $\Rightarrow$ diagonalisable in $\mathbb{C}$ (corollary).

Two possible obstacles: roots missing in $\mathbb{K}$ (Example 3) or $m_g(\lambda) < m_a(\lambda)$ (Example 2).

What it is for: powers of a matrix

Example 1: $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} = PDP^{-1}$, $P = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$, $D = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$.

What it is for: powers of a matrix

Example 1: $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} = PDP^{-1}$, $P = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$, $D = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$.

$$A^k = P \begin{bmatrix} 1 & 0 \\ 0 & 3^k \end{bmatrix} P^{-1} = \frac{1}{2} \begin{bmatrix} 3^k + 1 & 3^k - 1 \\ 3^k - 1 & 3^k + 1 \end{bmatrix}.$$

What it is for: powers of a matrix

Example 1: $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} = PDP^{-1}$, $P = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$, $D = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$.

$$A^k = P \begin{bmatrix} 1 & 0 \\ 0 & 3^k \end{bmatrix} P^{-1} = \frac{1}{2} \begin{bmatrix} 3^k + 1 & 3^k - 1 \\ 3^k - 1 & 3^k + 1 \end{bmatrix}.$$

For example, $A^5 = \begin{bmatrix} 122 & 121 \\ 121 & 122 \end{bmatrix}$: one calculation, instead of four matrix products.

Check: $k = 1$ gives A ; $k = 0$ gives I_2 .

Link to Ch. 5: symmetric matrices

Extension (not required in any exercise of Sheet 6).

In Example 1, $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ is **symmetric** and the eigenvectors $(1, -1)$ and $(1, 1)$ are **orthogonal** (Ch. 5).

Link to Ch. 5: symmetric matrices

Extension (not required in any exercise of Sheet 6).

In Example 1, $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ is **symmetric** and the eigenvectors $(1, -1)$ and $(1, 1)$ are **orthogonal** (Ch. 5). Normalising them gives an orthonormal basis of eigenvectors:

$$Q = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}, \quad Q^T Q = I_2, \quad A = Q D Q^T \quad \text{with } D = \text{diag}(1, 3).$$

Q is orthogonal (Ch. 3): $Q^{-1} = Q^T$.

Link to Ch. 5: symmetric matrices

Extension (not required in any exercise of Sheet 6).

In Example 1, $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ is **symmetric** and the eigenvectors $(1, -1)$ and $(1, 1)$ are **orthogonal** (Ch. 5). Normalising them gives an orthonormal basis of eigenvectors:

$$Q = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}, \quad Q^T Q = I_2, \quad A = Q D Q^T \quad \text{with } D = \text{diag}(1, 3).$$

Q is orthogonal (Ch. 3): $Q^{-1} = Q^T$.

Theorem (without proof)

If A is real and symmetric, its eigenvalues are real and there is an orthonormal basis of $\mathbb{R}^n$ made up of eigenvectors of A . In particular, A is diagonalisable: $A = Q D Q^T$, with Q orthogonal.

It underlies principal component analysis and the classification of minima using the Hessian.

Every year, 10 % of the city population moves to the countryside and 20 % of the countryside population moves to the city. With $\mathbf{x}_k = (\text{city}, \text{countryside})$ as fractions of the population:

$$\mathbf{x}_{k+1} = M\mathbf{x}_k, \quad M = \begin{bmatrix} 0.9 & 0.2 \\ 0.1 & 0.8 \end{bmatrix}.$$

Eigenvalues 1 and 0.7; eigenvectors $(2, 1)$ and $(1, -1)$. Starting from $\mathbf{x}_0 = (0.5, 0.5)$:

LINEAR ALGEBRA IN PRACTICE: long-term behaviour

Every year, 10 % of the city population moves to the countryside and 20 % of the countryside population moves to the city. With $\mathbf{x}_k = (\text{city}, \text{countryside})$ as fractions of the population:

$$\mathbf{x}_{k+1} = M\mathbf{x}_k, \quad M = \begin{bmatrix} 0.9 & 0.2 \\ 0.1 & 0.8 \end{bmatrix}.$$

Eigenvalues 1 and 0.7; eigenvectors $(2, 1)$ and $(1, -1)$. Starting from $\mathbf{x}_0 = (0.5, 0.5)$:

$$\mathbf{x}_k = \frac{1}{3} \begin{bmatrix} 2 \\ 1 \end{bmatrix} - \frac{1}{6} (0.7)^k \begin{bmatrix} 1 \\ -1 \end{bmatrix} \xrightarrow[k \rightarrow \infty]{} \begin{bmatrix} 2/3 \\ 1/3 \end{bmatrix}.$$

After 10 years, $\mathbf{x}_{10} \approx (0.662, 0.338)$.

The eigenvector for $\lambda = 1$ gives the equilibrium; $|0.7| < 1$ says that the other part dies out. The same idea is behind Markov chains and PageRank.

Check whether the following matrices are diagonalisable. For those that are, find matrices P and D such that $P^{-1}AP = D$.

$$100) \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 2 & 3 \end{bmatrix}$$

$$102 \text{ b)} \begin{bmatrix} 2 & 2 & -2 \\ 0 & 5 & -3 \\ 0 & 1 & 1 \end{bmatrix}$$

$$102 \text{ f)} \begin{bmatrix} -1 & 3 & -1 \\ -4 & 7 & -2 \\ -8 & 10 & -2 \end{bmatrix}$$

$$102 \text{ c)} \begin{bmatrix} 0 & -2 & -4 \\ 0 & 3 & 5 \\ -2 & -3 & -3 \end{bmatrix}$$

Solutions on the next slide. The rest are on Sheet 6: 102 a), d), e), g), h).

Ex. 100 and 102 — solutions

100) $\lambda = 1$ ($m_a = m_g = 2$), $\lambda = 4$. Diagonalisable: $P = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 2 & 0 & -1 \end{bmatrix}$, $D = \text{diag}(4, 1, 1)$.

102 b) $\lambda = 2$ ($m_a = m_g = 2$), $\lambda = 4$. Diagonalisable: $P = \begin{bmatrix} 2 & 1 & 0 \\ 3 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}$, $D = \text{diag}(4, 2, 2)$.

102 f) $\lambda = 1$ ($m_a = 2$, $m_g = 1$), $\lambda = 2$. **Not diagonalisable**: $m_g(1) < m_a(1)$.

102 c) $p(\lambda) = -(\lambda + 2)(\lambda^2 - 2\lambda + 2)$: $\lambda = -2$ and $\lambda = 1 \pm i$.

Not diagonalisable in $\mathbb{R}$; diagonalisable in $\mathbb{C}$ (3 distinct eigenvalues).

P is not unique: any choice of lin. indep. eigenvectors, in the order of D , will do.

LINEAR ALGEBRA ON THE COMPUTER: eigenvalues and the long term

For illustration only — not examinable. Python (NumPy); in MATLAB/Octave: $[P,D] = \text{eig}(C)$.

```
import numpy as np
C = np.array([[ 2, 0, 0],
              [-6, 2, -3],
              [-6, 0, -1]])
lam, P = np.linalg.eig(C)
lam    # [ 2. -1.  2.]
P      # columns: (0, 1, 0),
          # (0, 0.707, 0.707), (0.447, 0, -0.894)
M = np.array([[0.9, 0.2], [0.1, 0.8]])
x = np.array([0.5, 0.5])
for k in range(50): x = M @ x
x      # [0.6667 0.3333]
```

Example 4. The eigenvalues are $2, 2, -1$, in a different order. The columns of P are eigenvectors **of norm 1**: $(0, 1, 0)$, $\frac{1}{\sqrt{2}}(0, 1, 1)$ and $\frac{1}{\sqrt{5}}(1, 0, -2)$, the same as by hand, up to a factor. P is not unique: the scale and order change, but $P^{-1}CP$ is still diagonal.

City and countryside. Applying M repeatedly takes $\mathbf{x}_k$ to the eigenvector for $\lambda = 1$, $(\frac{2}{3}, \frac{1}{3})$: the part with 0.7^k dies out. This is the *power method*, the basic idea behind PageRank.

Concept	How to obtain it / what it says
Eigenvalue λ	root of $p(\lambda) = \det(A - \lambda I_n)$ in $\mathbb{K}$
Eigenvector	<i>non-zero</i> solution of $(A - \lambda I_n)\mathbf{v} = \mathbf{0}$
Eigenspace $E(\lambda)$	all solutions of $(A - \lambda I_n)\mathbf{v} = \mathbf{0}$ (includes $\mathbf{0}$)
$m_a(\lambda)$	multiplicity of λ as a root of p
$m_g(\lambda)$	$\dim E(\lambda) = n - \text{car}(A - \lambda I_n)$; $1 \leq m_g \leq m_a$
Linear operator T	matrix with columns $T(\mathbf{e}_j)$; same eigenvalues in any basis
Diagonalisable	there is an invertible P with $P^{-1}AP$ diagonal $\iff \sum m_g(\lambda) = n$
P and D	columns of P : lin. indep. eigenvectors; diagonal of D : eigenvalues, same order
Sufficient (not nec.)	n distinct eigenvalues $\Rightarrow$ diagonalisable
Triangular	eigenvalues = entries on the main diagonal
Checks	$\sum \lambda_i = \text{tr } A$, $\prod \lambda_i = \det A$ (in $\mathbb{C}$); $A\mathbf{v} = \lambda\mathbf{v}$; $AP = PD$
Powers	$A^k = PD^kP^{-1}$

KEY TAKEAWAYS — Chapter 6

1. An eigenvector gives a direction that the matrix does not change: the vector is only multiplied by a number, the eigenvalue.
2. Finding eigenvalues means asking when a homogeneous system has non-zero solutions, i.e. when a determinant vanishes.
3. An eigenvalue has two multiplicities, and the geometric one never exceeds the algebraic one.
4. Diagonalising means choosing a basis of eigenvectors; it is only possible when there are enough lin. indep. eigenvectors to form such a basis.
5. Distinct eigenvalues guarantee diagonalisation, but repeated ones do not prevent it.

Next (beyond this course): eigenvalues come back in differential equations, in optimisation (the Hessian matrix) and in statistics (principal components).

References (full list: course bibliography)

- Santana, A. P., Queiró, J. F., *Introdução à Álgebra Linear*, Gradiva (in Portuguese) — chapter on eigenvalues and eigenvectors.
- Lipschutz, S., Lipson, M., *Schaum's Outline of Linear Algebra*, McGraw-Hill — many worked exercises.
- Strang, G., *Introduction to Linear Algebra*, Wellesley-Cambridge Press — chapter *Eigenvalues and Eigenvectors*.
- Magalhães, L. T., *Álgebra Linear*, IST.

Resources — Sheet 6 (exercises 99–102) and its solutions.